

Co-3 Assessment Tool-3

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Course Code: -MLA0305

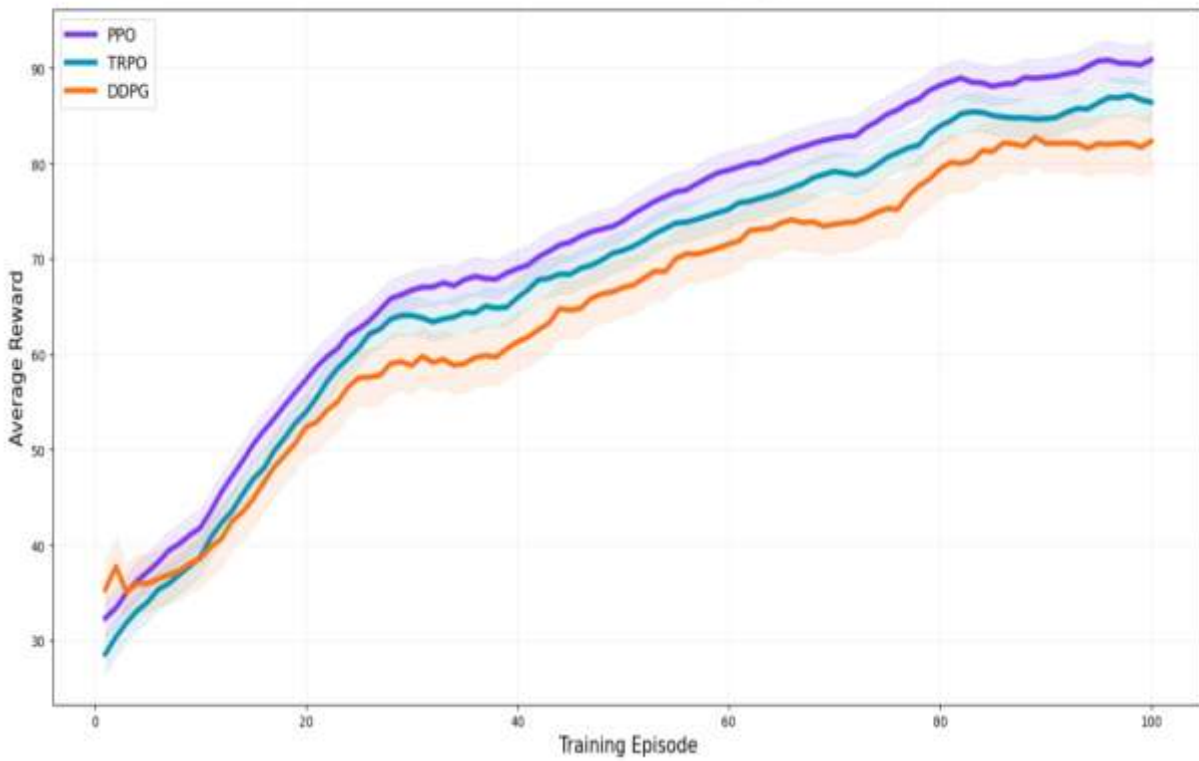
Slot: -B

PPO, TRPO & DDPG: Comprehensive Performance Analysis and Comparative Evaluation

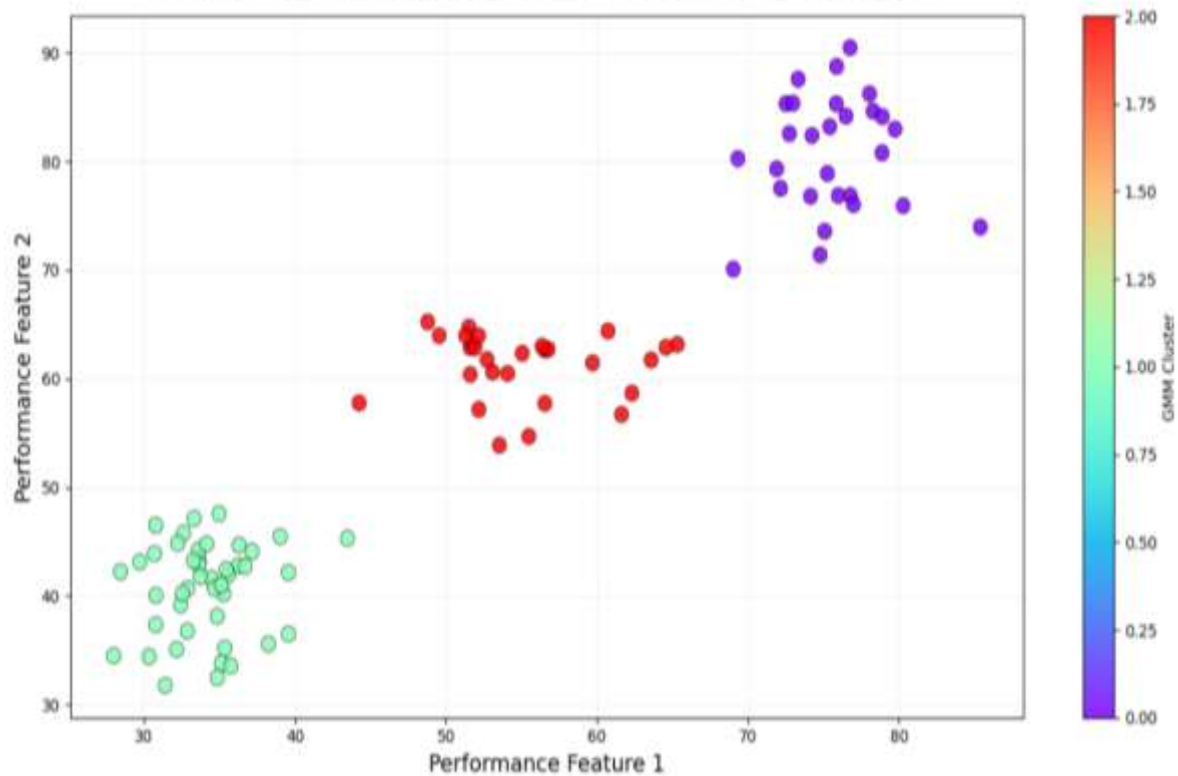
Episode	PPO_Reward	TRPO_Reward	DDPG_Reward	PPO_Average	TRPO_Average	DDPG_Average	GMM_Cluster
1	32.33	28.58	35.34	32.33	28.58	35.34	1
2	34.43	32.15	40.16	33.38	30.37	37.75	1
3	38.20	34.77	29.74	34.99	31.83	35.08	1
4	40.01	36.81	38.52	36.24	33.08	35.94	1
5	40.88	37.71	36.14	37.17	34.00	35.98	1
6	43.26	41.86	38.72	38.19	35.31	36.43	1
7	46.68	39.79	39.48	39.40	35.95	36.87	1
8	45.26	43.47	40.35	40.13	36.89	37.30	1
9	48.33	45.03	44.33	41.04	37.80	38.09	1
10	48.77	47.51	42.94	41.82	38.77	38.57	1

Fig1: - Sample Dataset

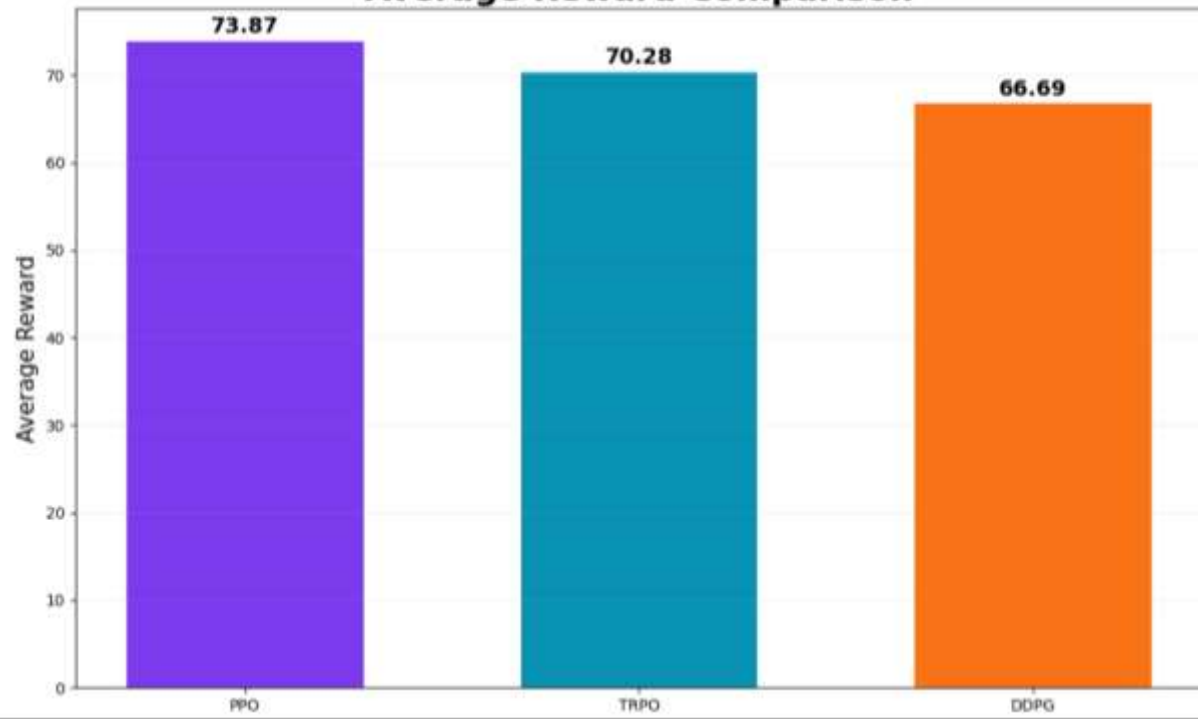
PPO vs TRPO vs DDPG — Learning Performance



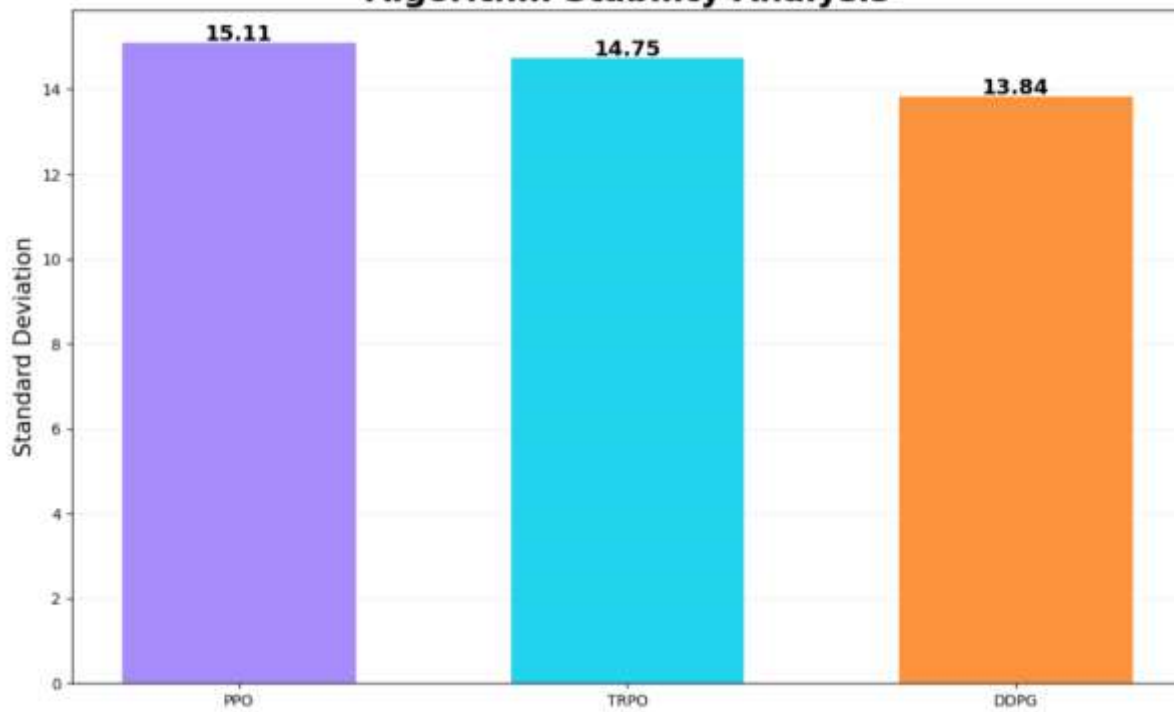
GMM-Based Synthetic Performance Dataset

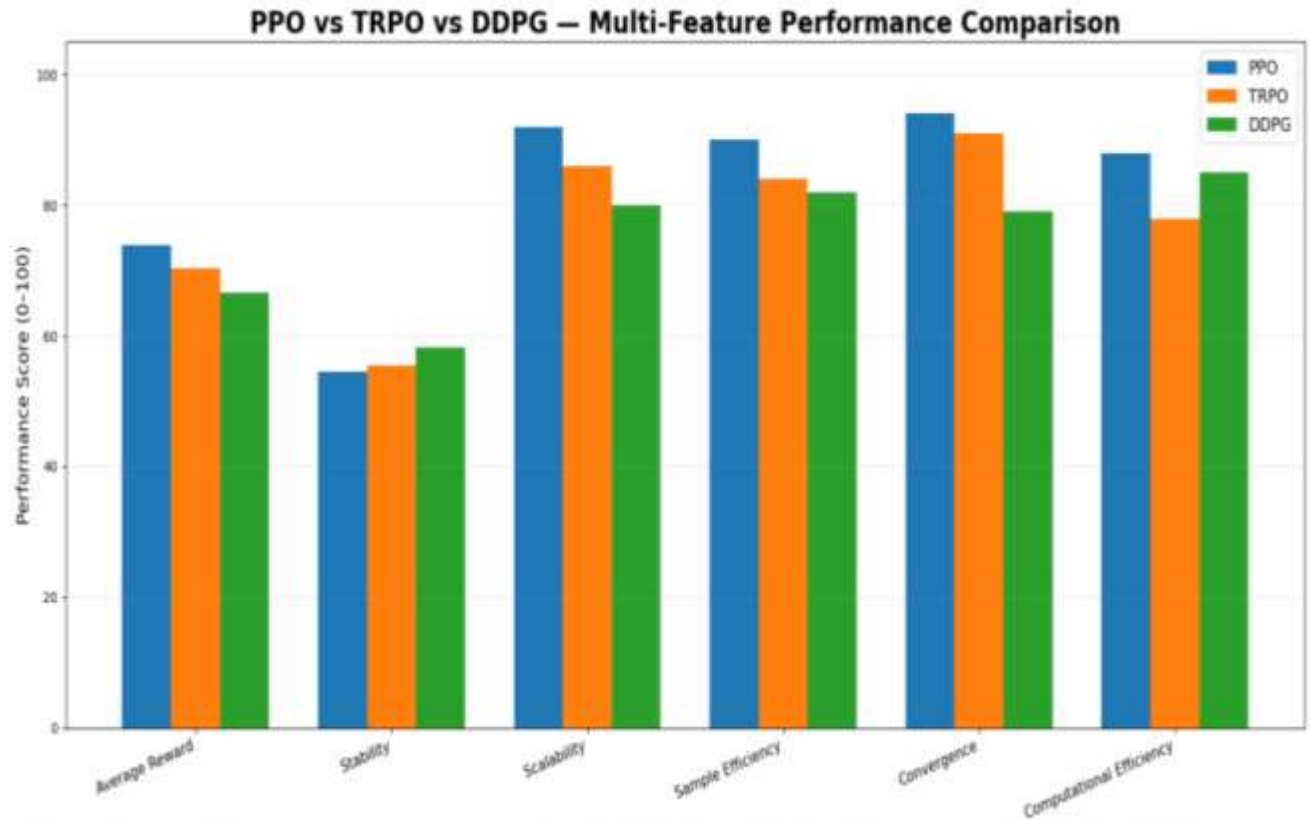


Average Reward Comparison



Algorithm Stability Analysis





Metric	PPO	TRPO	DDPG
Average Reward	73.87	70.28	66.69
Final 20-Episode Reward	89.93	85.53	82.20
Best Reward	94.22	90.82	89.14
Stability (Std. Dev.)	15.11	14.75	13.84
Overall Rank	1	2	3

Fig: - Summary Table

PPO vs TRPO vs DDPG — Overall Performance Profile

